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## **MI-6002 (CBGS)**

### **B.E. VI Semester**

Examination, December 2020

### **Choice Based Grading System (CBGS)**

### **Mining Environment - II**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Compare Boundary ventilation system with central ventilation system?  
b) Explain the air requirements for an underground mine for ventilation planning.
2. Discuss the series and parallel operation of the mine fan in detail.
3. Enumerate types of mine fans with details of their characteristics and suitability.
4. Derive the expression for theoretical head of centrifugal fan.
5. Write notes on the following (any two)
  - a) What are the physiological properties of dust? Explain.
  - b) Explain any one method of sampling an air-borne dust in mines.
  - c) What is Silicosis? Explain in detail.

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6. Write notes on the following (any two)
  - a) Explain the method of wet suppression of dust.
  - b) Explain the parallel operation of fans.
  - c) Explain the Hardy-Cross method of Ventilation Network Analysis.
7. Describe various instruments used for measurement of pressure, velocity and quantity of air.
8. List out the different types of dust sampler use in mine with their working principle.

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